Worksheet 1-16 A Intermediate Value Theorem #1
AP Calculus

Name SOLUTIONS

Directions: Answer the following problems. Show all work and use proper mathematics when using theorems.

x	1	3	4	7	8
$f(x)$	3	-1	5	2	9

1. Let $f(x)$ be a continuous function with selected values shown in the table above. Must there be a value c for $1 < c < 8$ such that $f(c) = 8$? Give a reason for your answer.

Yes, since f is continuous on the closed interval [1,8], where $f(1) \neq f(8)$ and 8 is a number between $f(1) = 3$ and $f(8) = 9$, then the IVT guarantees there is at least one number c in (1,8) such that $f(c) = 8$.

t	0	1	5	8	10
$v(t)$	24.3	18.1	11.7	3.1	-4.2

2. A particle is moving along a horizontal axis. Let $v(t)$ be a continuous function for the velocity of the particle, in meters per second, at time t seconds. Is there a time r, $0 < r < 10$ such that $v(r) = 0$? Justify your answer.

Yes, since $v(t)$ is continuous on the closed interval [0,10], where $v(0) \neq v(10)$ and 0 is a number between $v(0) = 24.3$ and $v(10) = -4.2$, then the IVT guarantees there is at least one time r in (0,10) such that $v(r) = 0$.

t (months)	1	2	5	9	12
$H(t)$ (feet)	5	5.7	7.1	11.8	14.5

3. After a tree is planted, the height of the tree, in feet, is measured at different times over the course of the year. For $0 \leq t \leq 12$ months, the height of the tree can be modeled by the continuous function $H(t)$, measured in feet. For $1 < t < 12$, must there be a time when $H(t) = 10$? Give a reason for your answer.

Yes, since $H(t)$ is continuous on the closed interval [1,12], where $H(1) \neq H(12)$ and 10 is a number between $H(1) = 5$ and $H(12) = 14.5$, then the IVT guarantees there is at least one time in (1,12) such that $H(t) = 10$.

x	-3	0	1	5	9
$g(x)$	2	13	19	11	12

4. Let $g(x)$ be the continuous function with selected values shown in the table above. Show that there must be a value k for $-3 < k < 9$ such that $g(k) = 15$.

$g(x)$ is continuous on the closed interval $[-3,1]$, which is in the closed interval $[-3,9]$, where $g(-3) \neq g(1)$ and 15 is a number between $g(-3) = 2$ and $g(1) = 19$, then the IVT guarantees there is at least one value k in $(-3,1)$, which is in the interval $(-3,9)$, such that $g(k) = 15$.

x	$f(x)$	$g(x)$
1	6	2
2	9	3
3	10	4
6	-1	6
10	3	11

5. The functions f and g are continuous for all real numbers, and g is strictly increasing. The table above gives values of the functions at selected values of x. The function h is given by $h(x) = g(f(x))$.

Explain why there must be a value r for $1 < r < 3$ such that $h(r) = 8$.

$h(x)$ is continuous because f and g are continuous.

$$h(1) = g(f(1)) = g(6) = 6 \quad \text{and} \quad h(3) = g(f(3)) = g(10) = 11.$$

$h(x)$ is continuous on the closed interval $[1,3]$, where $h(1) \neq h(3)$ and 8 is a number between $h(1) = 6$ and $h(3) = 11$, then the IVT guarantees there is at least one value r in $(1,3)$ such that $h(r) = 8$.

x	0	4	6	8	13
$f(x)$	3	4.5	3	2.5	4.4

4 4 4

6. The table above shows selected values of a continuous function f. For $0 \leq x \leq 13$, what is the fewest possible number of times that $f(x) = 4$?

(A) One (B) Two (C) Three (D) Four

x	-1	3	5	6	10
$f(x)$	2	0	1	-1	5

7. The table above gives selected values for the continuous function, $f(x)$. What is the minimum number of zeros guaranteed on $f(x)$? Explain your reasoning.

There are at least three zeros. There is already one at $x = 3$. The IVT guarantees there is at least one between $x = 5$ and $x = 6$ and another one between $x = 6$ and $x = 10$ because $f(x)$ changes signs between these x–values.

x	-1	2	3	6
$h(x)$	1	k	2	-5

8. The table above gives selected values for the continuous function, $h(x)$.

a) If $h(x)$ has exactly 2 zeros, what is the value of k that guarantees $h(x)$ has two zeros?

There is a zero between $x = 3$ and $x = 6$ because $h(x)$ changes signs on this interval. $k = 0$ will then guarantee only one zero between $x = -1$ and $x = 3$.

b) If $h(x)$ only has 1 zero, what is one possible value of k?

Answers will vary but $k > 0$.

c) What is one possible value of k that guarantees $h(x)$ will have at least 3 zeros?

Answers will vary but $k < 0$. When $k < 0$ then there will be a zero between $x = -1$ and $x = 2$ then another one between $x = 2$ and $x = 3$.

9. Let g be a function such that $g(-1) = 0$ and $g(2) = 5$. Which of the following conditions guarantees that there is an x, $-1 < x < 2$, $g(x) = 3$?

(A) g is defined for all x in $(-1, 2)$
(B) g is continuous for all x in $[-1, 2]$
(C) g is increasing on $[-1, 2]$
(D) there exists an x in $(-1, 2)$ such that $g(x) = 6$

10. Let f be a function that is continuous on the closed interval [2,4] with $f(2) = 10$ and $f(4) = 20$. Which of the following is guaranteed by the Intermediate Value Theorem?

(A) $f(x) = 13$ has at least one solution in the open interval (2,4)
(B) $f(3) = 15$
(C) f attains a maximum on the open interval (2,4)
(D) $f(x) = 10$ at some other value(s)of x other than $x = 2$

11. Let f be a function that is continuous on the closed interval $[1, 3]$ with $f(1) = 10$ and $f(3) = 18$. Which of the following statements must be true?

(A) $10 \leq f(x) \leq 18$
(B) f is increasing on the interval $[1, 3]$
(C) $f(x) = 17$ has at least one solution in the interval $[1, 3]$
(D) $f(2) = 14$

12. Let f be a function such that $f(1) = -2$ and $f(5) = 7$. Which of the following conditions guarantees that $f(c) = 0$ for some value c in the open interval $(1, 5)$?

(A) f is increasing on the closed interval $[1, 5]$
(B) f is continuous on the closed interval $[1, 5]$
(C) $f(3) = 3.5$
(D) f is decreasing on the closed interval $[1, 5]$